

PHARMACY MANAGEMENT SOFTWARE

Corrected Final Portal Structure + HMS Integration

Management Portal (Super Admin/Admin) + Sales Portal

Corrected Architecture

Super Admin and Admin do **NOT** have separate operational portals. Both use the same Management Portal and the same pharmacy screens. Their difference is permission-based: Super Admin creates/manages Admin accounts; Admin creates/manages Sales accounts. Protected Super Admin access cannot be changed by Admin.

Audit Rule

Every POS sale, HMS fulfillment, purchase, vendor payment, expense, return, stock adjustment, petty cash transaction and settlement must store and display the actual user who performed it. Reports show User Name + Role + Source Portal/Module + Date/Time.

1. Final Portal Architecture

Portal	Who Uses It	Main Difference
Management Portal	Super Admin + Admin	Same operational screens. Super Admin manages Admin accounts and has protected/root oversight. Admin manages Sales users and normal pharmacy operations.
Sales Portal	Sales / Cashier staff	POS, HMS request fulfillment, allowed returns, receipts, own Balance Sheet, own Account Settlement and own reports.

1.1 Role Difference Inside Same Management Portal

Capability	Super Admin	Admin
Dashboard / POS / Stock / Purchase / Vendors	Yes	Yes
Vendor Ledger / Payments	Yes	Yes
Expenses / Petty Cash / Settlements	Yes	Yes
Operational Reports	Yes - full scope	Yes - normal management scope
Create Admin Account	Yes	No
Create Sales Account	No - Admin does this	Yes
Change Protected Super Admin	Protected/root governance only	No
Security / Admin Account Audit	Full	Normal operational visibility only

UI Recommendation

Do not build two different Management Portal UIs. Render the same screens and hide/disable only role-restricted actions. Example: Super Admin sees '+ Add Admin'; Admin sees '+ Add Sales User'.

2. Management Portal - Final Sidebar

Group	Screens / Tabs	Purpose
Dashboard	Pharmacy Dashboard	Sales, profit, purchases, stock, expiry, vendor payable, cash and settlement overview.
POS & Sales	Sales Overview; Returns/Voids	Monitor sales, payments, discounts and controlled reversals.
HMS Requests	Hospital Request Queue	Receive and fulfill medicine requests coming from Hospital Management System.
Medicines & Stock	Medicine Stock Overview; Batch/Expiry; Stock Movement	Medicine master, live batches, expiry, low/out stock and audited movements.
Purchases	Purchase / Stock In; Purchase Return	Receive medicines from vendors and return purchase stock.
Vendors	Vendor Directory; Vendor Ledger	Vendor master plus separate financial running ledger.
Cash & Expenses	Petty Cash; Expenses; Balance Sheet; Account Settlement	Cash control and operating expenses.
Users	Role-aware Account Management	Super Admin: Admin accounts. Admin: Sales accounts.
Reports	Pharmacy Reports	One consolidated reporting center.
Settings	Pharmacy Settings	Tax/discount, receipt, stock/expiry, high-value approval, policies.

3. Management Dashboard

Dashboard should quickly show what is selling, what needs stock attention, what money is due and which settlements need action.

KPI Card	Meaning	Click Through
Sales Today	Net pharmacy sales today.	Sales Overview.
Gross Profit Today	Net sales minus cost of sold medicines.	Sales/Profit report.
Purchases Today	Posted purchase count and value.	Purchases.
Current Stock Value	Valuation of available pharmacy stock.	Stock Overview.
Low Stock	Medicines at/below reorder level.	Filtered Stock.
Out of Stock	Medicines with zero available quantity.	Filtered Stock.
Near Expiry	Batches within configured expiry window.	Batch/Expiry.
Expired / Quarantine	Blocked/non-saleable batches.	Batch/Expiry.
Vendor Payable	Current total payable to vendors.	Vendor Ledger.
Pending HMS Requests	Hospital requests waiting for action.	HMS Request Queue.
Pending Settlements	Sales/Admin settlements awaiting review.	Account Settlement.
Expected Cash	Cash position for current logged-in cash handler.	Balance Sheet.

Dashboard Quick Actions

Button	Action
+ New Purchase	Open Purchase / Stock In.
+ Add Medicine	Open Medicine Master.
+ Add Vendor	Open Vendor form.
Pay Vendor	Open Vendor Ledger payment.
HMS Requests	Open pending hospital requests.
+ Expense	Create pharmacy expense.
Issue Petty Cash	Admin issues petty/opening cash to Sales user.
My Balance Sheet	Open current user's cash position.

4. Sales Portal - Keep It Focused

Group	Screens	Purpose
Dashboard	Sales Dashboard	My sales, cash/card/online, HMS requests, returns, expected cash.
POS	New Sale	Direct pharmacy sale.
HMS Requests	Request Queue	Fulfill hospital medicine requests.
Returns	Sales Return	Controlled return/refund against original sale.
Cash Control	My Balance Sheet; My Account Settlement	Own cash and handover.
Reports	My Sales Reports	Own sales, collections, returns and settlement reports.

Sales User Restriction

Sales user cannot create users, vendors, medicine masters, purchase stock or manually change quantity. Sales works through POS, HMS fulfillment, authorized returns and personal cash control.

4.1 Sales Dashboard

Card / Panel	What it shows
My Sales Today	Own net sales.
Cash Collection	Physical cash collected.
Card / Online	Non-cash collection.
Returns / Refunds	Own approved returns/refunds.
Pending HMS Requests	Medicine requests waiting to be fulfilled.
My Expected Cash	System expected physical cash.
Low / Out Stock Alerts	Read-only stock alerts.
Recent Transactions	Latest sale/request/refund activity.

5. POS - Direct Pharmacy Sale

Area	Working
Customer / Patient	Optional direct customer or linked HMS patient/admission reference.
Search / Barcode	Search by medicine name/code or scan barcode.
Batch	System selects valid FEFO batch; authorized override requires reason.
Sale Line	Medicine, batch, expiry, qty, sale rate, discount, tax if enabled, line net.
Validation	Cannot sell above available qty. Expired/quarantined stock blocked.
Totals	Gross -> Discount -> Tax -> Net Payable.
Payment	Cash / Card-POS / Online-Bank; split payment only if enabled.
Receipt	Invoice/Receipt no, cashier, patient/customer, lines, totals, method, date/time.
Buttons	Hold Sale Complete Sale Complete & Print Clear Cart.

5.1 Direct POS Amount Calculation

Calculation	Formula
Line Gross	Qty x Sale Price.
Line Discount	Fixed or percentage if user/policy permits.
Line Net Before Tax	Line Gross - Line Discount.
Line Tax	Configured tax on taxable amount.
Invoice Net	Sum of line net amounts + tax - invoice-level approved discount, if enabled.
Paid	Sum of valid Cash/Card/Online payments.
Outstanding	Invoice Net - Valid Paid + Valid Refund/Reversal effect.

6. Hospital Management System Integration

Integration Boundary

Hospital and Pharmacy remain separate systems/financial streams. HMS sends medicine requests; Pharmacy fulfills from its own stock and produces a separate Pharmacy invoice/clearance status. Pharmacy does not merge its bill into the Hospital Bill.

6.1 HMS -> Pharmacy Request Flow

Step	HMS / Hospital Side	Pharmacy Side
1	Admission/authorized clinical workflow creates medicine request.	Request appears as Pending HMS Request.
2	Sends Admission/Encounter Ref, patient snapshot, medicine, qty, urgency, requester.	Validate medicine, batch, expiry and stock.
3	If high-value rule applies, request carries/awaits required approval.	Block dispense until configured approval is satisfied.
4	Waits for response.	Accept Full / Accept Partial / Reject with reason.
5	-	Dispense actual qty; post Stock OUT from exact batch.
6	Receives status and dispensed qty.	Create/update separate Pharmacy Invoice.
7	Receives Pharmacy amount/due/clearance summary.	Collect Pharmacy payment or mark approved credit/panel status according to policy.
8	Admission sees Pharmacy clearance separately from Hospital clearance.	Return final request/invoice/payment/clearance status to HMS.

7. HMS Request Queue - UI

Area	What appears
Top Filters	From/To, Status, Admission No, Patient, Medicine, Urgency, Approval Status.
Summary Cards	Pending, Approved to Dispense, Partial, Fulfilled, Rejected, High-Value Approval Pending.
Table	Request Ref, Admission No, Patient, Medicine, Requested Qty, Dispensed Qty, Urgency, Requested By, Status, Age/Wait Time.
Row Actions	View Request Accept Partial Fulfill Reject Open Pharmacy Invoice.
Fulfillment Buttons	Accept & Dispense Partial Fulfill Reject Print Issue/Invoice.

7.1 High-Value Medicine Approval

Rule	Behaviour
Threshold	Configured by Management/Super Admin settings; not hardcoded.
Basis	Can be per medicine line or request/invoice amount, based on policy.
Applies To	Normal/Self-Pay and Panel cases as configured.
Approval	Attendant/management approval logic can be configured.
Fulfillment	Dispense is blocked while required approval is pending.
Audit	Requested By, Approved By, Approval Time and evidence/reference retained.

8. Pharmacy Invoice & Payment Calculation for HMS Requests

Separate Bill

For an inpatient/HMS request, Pharmacy creates a separate Pharmacy Invoice. Hospital Bill and Pharmacy Bill must remain separate. HMS receives only the linked Pharmacy invoice summary and clearance status required for discharge.

Item	Calculation / Rule
Dispensed Line Gross	Actual Dispensed Qty x Pharmacy Sale Price.
Line Discount	Only approved/configured pharmacy discount.
Tax	Configured pharmacy tax, if applicable.
Pharmacy Subtotal	Sum of valid dispensed line amounts.
Returns / Credits	Eligible approved medicine returns reduce Pharmacy amount.
Net Pharmacy Bill	Subtotal - Approved Discounts + Tax - Valid Credits/Returns.
Payments	Cash/Card/Online collected by Pharmacy Sales user.
Pharmacy Outstanding	Net Pharmacy Bill - Valid Pharmacy Payments.
Clearance	CLEARED when outstanding is zero, or APPROVED CREDIT/PANEL when authorized policy allows.

What HMS Receives Back

Request Status + Dispensed Qty + Pharmacy Invoice No + Pharmacy Net Amount + Paid Amount + Pharmacy Outstanding + Clearance Status + actual Pharmacy actor/user where relevant. HMS does not receive Pharmacy cashier ledger internals.

9. Medicines, Batches, Expiry & Stock

Area	Fields / Rules
Medicine Master	Code/Barcode, Name, Generic/Brand, Category, Unit/Pack, Reorder Level, Default Sale Price, Tax, Active Status.
Batch	Batch No, Expiry Date, Purchase Cost, Sale Price snapshot/default, Received Qty, Available Qty.
FEFO	Nearest valid expiry is issued/sold first.
Low Stock	Available Qty <= Reorder Level.
Out of Stock	Available Qty = 0.
Near Expiry	Within configured 30/60/90 or custom expiry window.
Expired	Cannot sell/dispense.
Quarantine	Not available for sale/dispense.
Quantity Rule	Never directly edit current qty; derive from posted stock movements.

9.1 Stock Movement Types

Movement	Effect
PURCHASE IN	Stock increases from posted purchase/GRN.
POS SALE OUT	Stock decreases on completed direct sale.
HMS DISPENSE OUT	Stock decreases on completed HMS fulfillment.
SALES RETURN IN	Eligible usable return increases stock.
PURCHASE RETURN OUT	Stock decreases when returned to vendor.
POSITIVE / NEGATIVE ADJUSTMENT	Approved count correction, damage, expiry write-off, loss, surplus, quarantine.
OPENING STOCK	Authorized migration/setup only.

10. Purchases & Vendor Management

10.1 Purchase / Stock In

Form Area	Fields / Buttons
Header	Vendor, Purchase Date, Vendor Invoice No, Payment Type: Cash/Bank/Credit, Notes.
Medicine Grid	Medicine, Batch, Expiry, Qty, Unit Cost, Discount/Tax, Line Total.
Finance	Purchase Total, Paid Now, Vendor Due.
Attachment	Vendor invoice/bill.
Buttons	Save Draft Post Purchase Post & Print GRN Cancel.

10.2 Vendor Directory

Area	Content
Buttons	+ Add Vendor Export Print
Filters	Search Active/Inactive Payment Terms
Table	Code, Name, Contact, Phone, Terms, Current Due, Status.
Actions	View Edit Open Ledger New Purchase Pay Vendor.

10.3 Vendor Ledger - Separate Financial Screen

Area	Content
Header	Vendor, contact, payment terms, Current Payable/Advance.
Buttons	Pay Vendor Purchase Return Approved Adjustment Print Statement PDF Excel.
Filters	From Date, To Date, Transaction Type.
Ledger	Date, Ref, Type, Description, Purchase, Payment, Return/Credit, Adjustment, Running Balance, Actor.
Totals	Opening + Purchases - Payments - Returns/Credits +/- Adjustments = Closing.

11. Petty Cash, Balance Sheet & Settlement

11.1 Petty Cash Flow

From	To	Effect
Super Admin / Management	Admin	Optional opening/petty cash for management operations.
Admin	Sales User	Opening/petty cash for Sales shift/POS.
Sales User	Admin	Physical cash handover through Account Settlement.
Admin	Management / Super Admin	Optional upper-level management settlement according to pharmacy policy.

11.2 Sales Balance Sheet

Item	Meaning
Opening / Carry	Approved opening cash.
Petty Cash Received	Cash received from Admin.
Cash POS Sales	Physical cash from direct pharmacy sales.
Cash HMS Payments	Physical cash collected against Pharmacy invoices for HMS requests.
Cash Refunds	Approved physical cash refunds.
Expected Cash	System-calculated amount.
Physical Cash	User counted amount.
Variance	Physical - Expected.
Settlement	Cash handed to Admin.
Remaining	Approved retained cash.

Sales Expected Cash

Opening/Carry + Petty Cash Received + Cash POS Sales + Cash HMS Payments + Approved Cash In - Cash Refunds - Previous Accepted Settlements +/- approved cash adjustments.

12. Expenses

Field	Rule
Category	Configurable expense category.
Date	Required.
Amount	Required.
Payment Method	Cash / Bank / Online.
Payee / Vendor	Optional.
Description / Reference	Required.
Attachment	Optional receipt.
Entered By	Actual logged-in user.
Approved By	If configured approval applies.
Buttons	Save Expense Save & Add Another Cancel.

Expense Audit

Reports must always show who entered the expense, who approved it (if approval exists), the source portal/module, payment method, date/time and reference. A cash expense affects the relevant user's expected physical cash.

13. User Management - Corrected

Logged-in Role	What account can be created?	Main Buttons
Super Admin	Admin accounts only	+ Add Admin Reset Password Activate/Deactivate View Last Login View Security Activity
Admin	Sales accounts only	+ Add Sales User Reset Password Activate/Deactivate Restrict/Enable Login View Last Login
Sales	None	No user management

14. Reporting - Actor Attribution is Mandatory

Every Report Row Must Answer: WHO DID IT?

A transaction must never appear as only 'Pharmacy' or 'Admin'. Show the actual user name. Recommended fields: Created By, Sold By, Collected By, Dispensed By, Entered By, Paid By, Refunded By, Adjusted By, Approved By, Settled By as applicable. Also include Role + Source Portal/Module + timestamp.

14.1 Consolidated Management Reports

Report	Main Data	Actor Fields
Sales & Collection	Invoice, patient/customer, gross, discount, net, method, paid/outstanding.	Sold By, Collected By.
HMS Request / Dispense	Request, admission, medicine, requested/dispensed qty, invoice, due/clearance.	Requested By from HMS, Dispensed By Pharmacy.
Purchase	GRN, vendor, batch, expiry, qty, cost, paid/due.	Received/Entered By, Paid By.
Stock Movement	Movement type, medicine, batch, qty in/out, balance.	Performed By, Approved By if adjustment.
Vendor Ledger	Purchase, payment, credit/return, running payable.	Created By / Paid By / Adjusted By.
Expense	Category, amount, method, description, status.	Entered By, Approved By.
Return / Refund	Original sale/purchase ref, qty/value, reason.	Processed/Refunded By, Approved By.
Balance / Settlement	Expected, physical, variance, submitted, accepted, remaining.	Submitted By, Reviewed/Accepted By.

14.2 Common Report Filters

Filter Group	Filters
Time	From Date, To Date, Today/This Month/Custom.
User	User Name, Role, Portal/Module.
Sales	Payment Method, Invoice Status, Return Status.
HMS	Admission No, Request Status, Approval Status, Clearance Status.
Stock	Medicine, Category, Batch, Stock Status, Expiry Window.
Vendor	Vendor, Transaction Type, Due Status.
Finance	Payment Method, Settlement Status, Expense Category.

15. HMS <-> Pharmacy Integration Data Contract

Direction	Data
HMS -> Pharmacy	Request Ref, Admission/Encounter Ref, patient display snapshot, Panel/Self-Pay indicator if required, medicine identifier, requested qty, urgency, requester, request timestamp, high-value approval reference/status when applicable.
Pharmacy -> HMS	Request status, accepted/rejected/partial state, actual dispensed qty, Pharmacy Invoice No, Pharmacy Net Amount, Pharmacy Paid Amount, Pharmacy Outstanding, Pharmacy Clearance Status, dispense timestamp, actual Dispensed By user.

HMS does NOT send Hospital cashier ledger, Hospital Balance Sheet, Hospital settlement internals.

Pharmacy does NOT send detailed vendor ledger, purchase cost ledger, full Pharmacy cashier balance sheet to clinical screens.

Source of Truth

HMS is source of truth for Admission/Encounter/request identity. Pharmacy is source of truth for medicine stock, batch/expiry, dispensed quantity, Pharmacy invoice, Pharmacy payments and Pharmacy clearance.

16. End-to-End Example

Step	Example
1	Admission sends request: Admission A-102, Ceftriaxone 1g x 2, Urgent.
2	Pharmacy queue shows Pending. Sales user opens request.
3	System finds valid FEFO batch: Batch CTX-45, Expiry 2027-03, Available 20.
4	If high-value approval is required, fulfillment waits for approval.
5	Sales dispenses Qty 2. Stock OUT = 2 from CTX-45.
6	Sale price PKR 900 each -> Gross PKR 1,800. Approved discount PKR 0. Tax PKR 0 -> Pharmacy Net = PKR 1,800.
7	Patient pays PKR 1,000 Cash -> Pharmacy Outstanding = PKR 800.
8	Sales user's Balance Sheet Cash HMS Payments increases by PKR 1,000.
9	Pharmacy returns to HMS: Dispensed 2, Invoice PH-0098, Net 1,800, Paid 1,000, Outstanding 800, Clearance = OUTSTANDING.
10	Later PKR 800 is paid -> Pharmacy Outstanding 0 -> Clearance = CLEARED -> HMS receives updated status.

17. Final Recommended Sidebar

Portal	Final Navigation
Management Portal - Super Admin/Admin	Dashboard POS & Sales HMS Requests Medicines & Stock Purchases Vendors Cash & Expenses Users Reports Settings
Sales Portal	Dashboard POS HMS Requests Returns Cash Control Reports

18. Final Non-Negotiable Rules

Rule	Decision
Super Admin/Admin architecture	Same Management Portal, role-based permissions.
User creation	Super Admin -> Admin only. Admin -> Sales only.
Actor attribution	Every action/report uses actual user name + role + portal/module + time.
HMS integration	Request in, fulfillment/invoice/clearance status out.
Financial separation	Hospital Bill and Pharmacy Bill remain separate.
Pharmacy payment	Collected/owned by Pharmacy; Pharmacy outstanding/clearance returned to HMS.
Batch/expiry	Mandatory where medicine is batch tracked; FEFO.
Expired stock	Blocked from sale/dispense.
Direct quantity edit	Not allowed.
Vendor Ledger	Separate dedicated financial ledger.
Cash control	User-wise Balance Sheet + Account Settlement.
Reports	Active filters + PDF/Excel/Print + actual actors.

Final Build Direction

Build one shared Management Portal for Super Admin/Admin with permission guards, plus one focused Sales Portal. Integrate HMS requests through stable request/invoice/status APIs. Keep stock, Pharmacy billing, Pharmacy cash and vendor accounting owned by the Pharmacy system.